

NirmanSathi

National Digital Construction Governance & Marketplace Platform for Nepal

SOFTWARE ARCHITECTURE

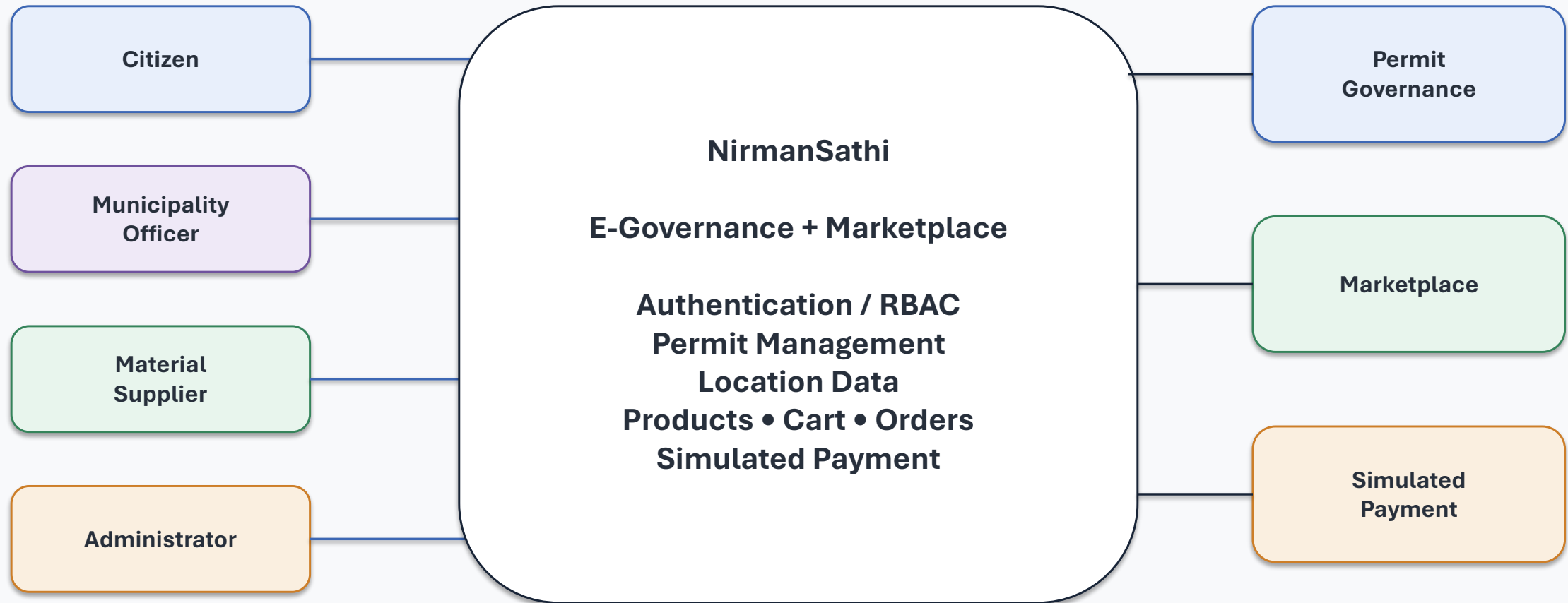
4+1 View Model • Current Implementation

BCS CSIT Project • Django + Bootstrap + JavaScript + SQLite

1. System Context

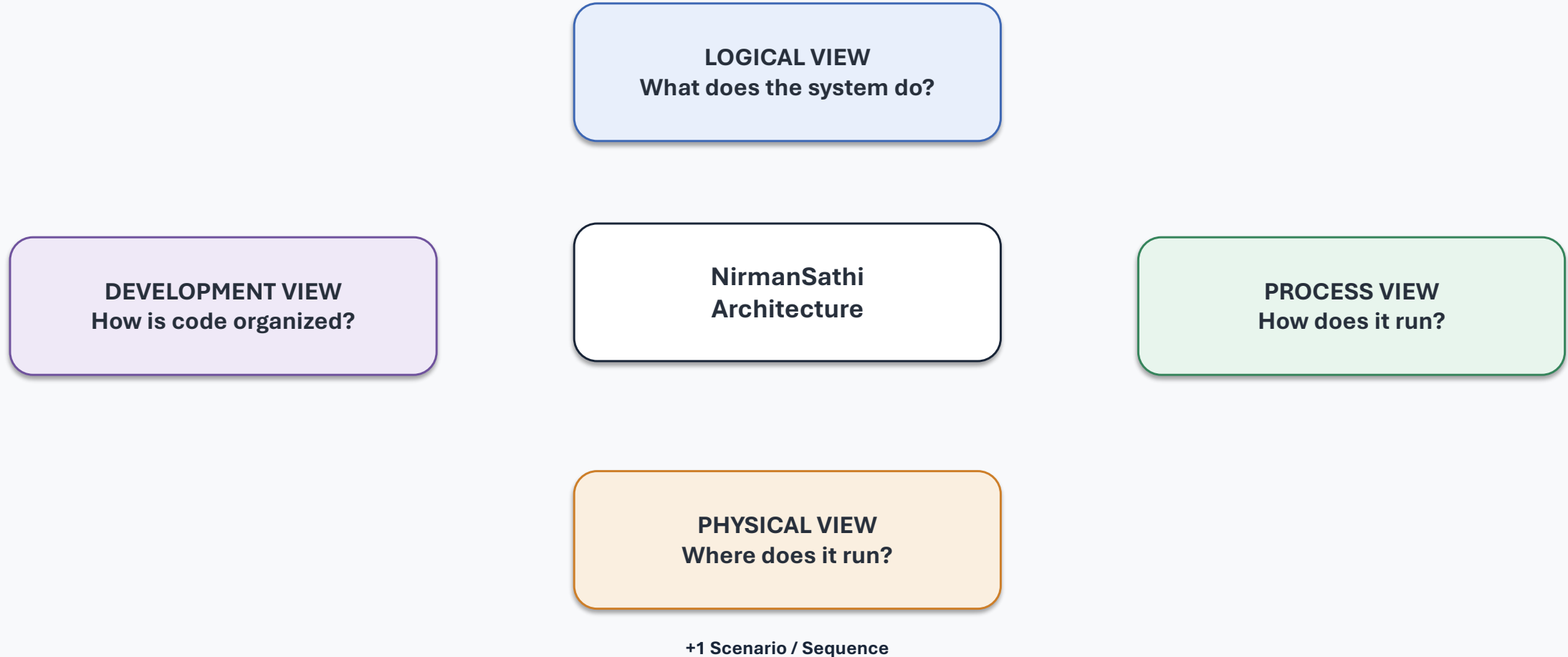
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Who uses NirmanSathi and what the platform provides



2. 4+1 Architecture Model

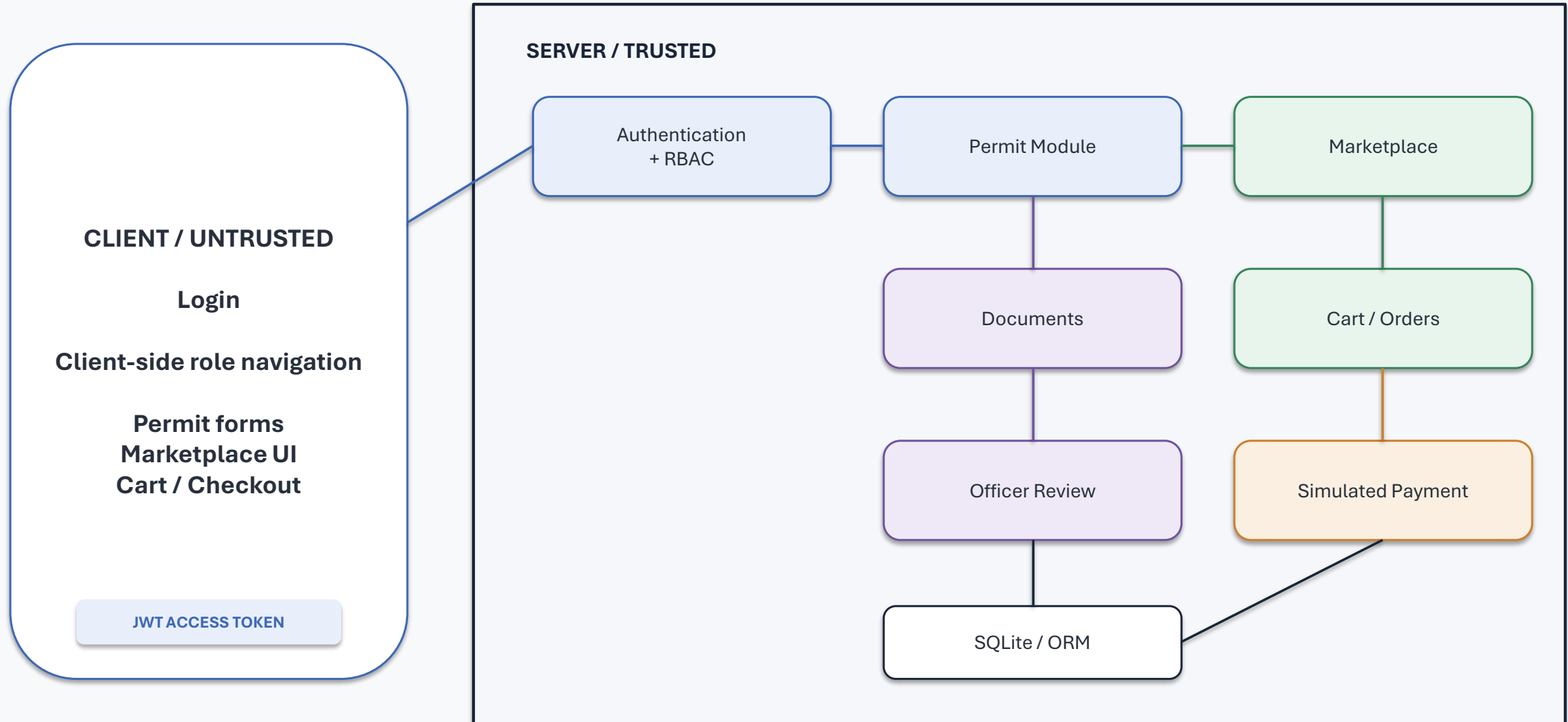
Each view answers a different architectural question



3. Logical View — Master Architecture

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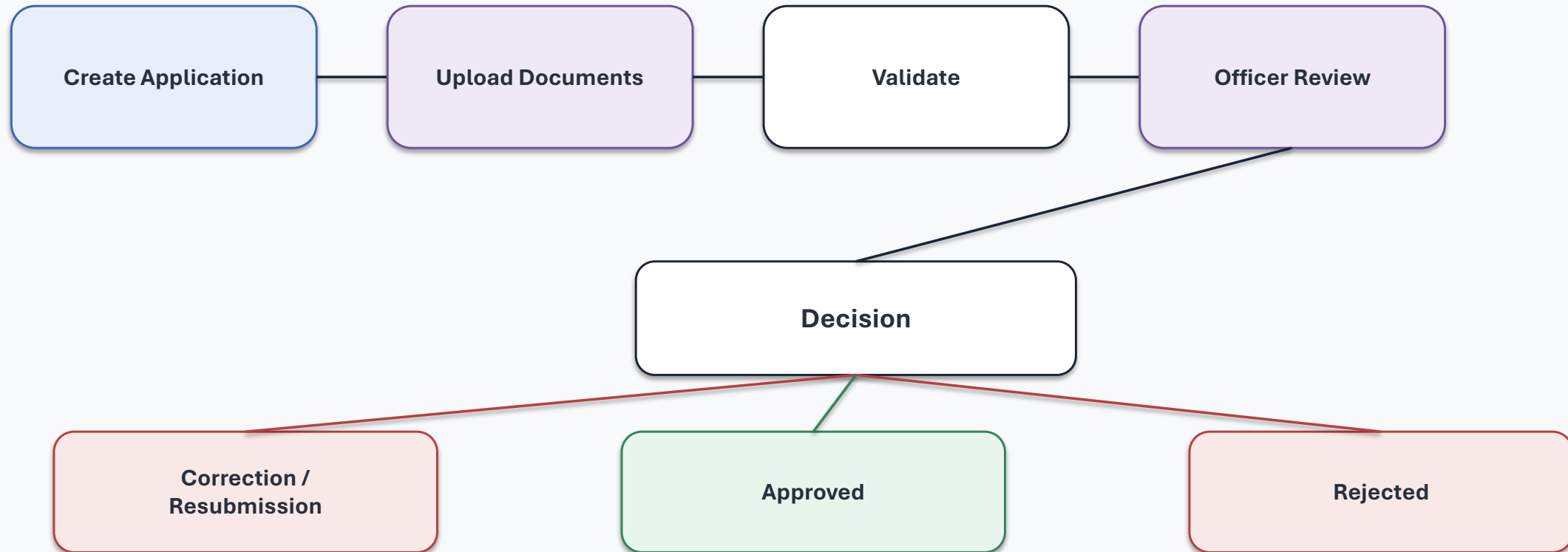
Business capabilities and the security boundary



4. Logical View — Permit Workflow

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Core e-governance business logic

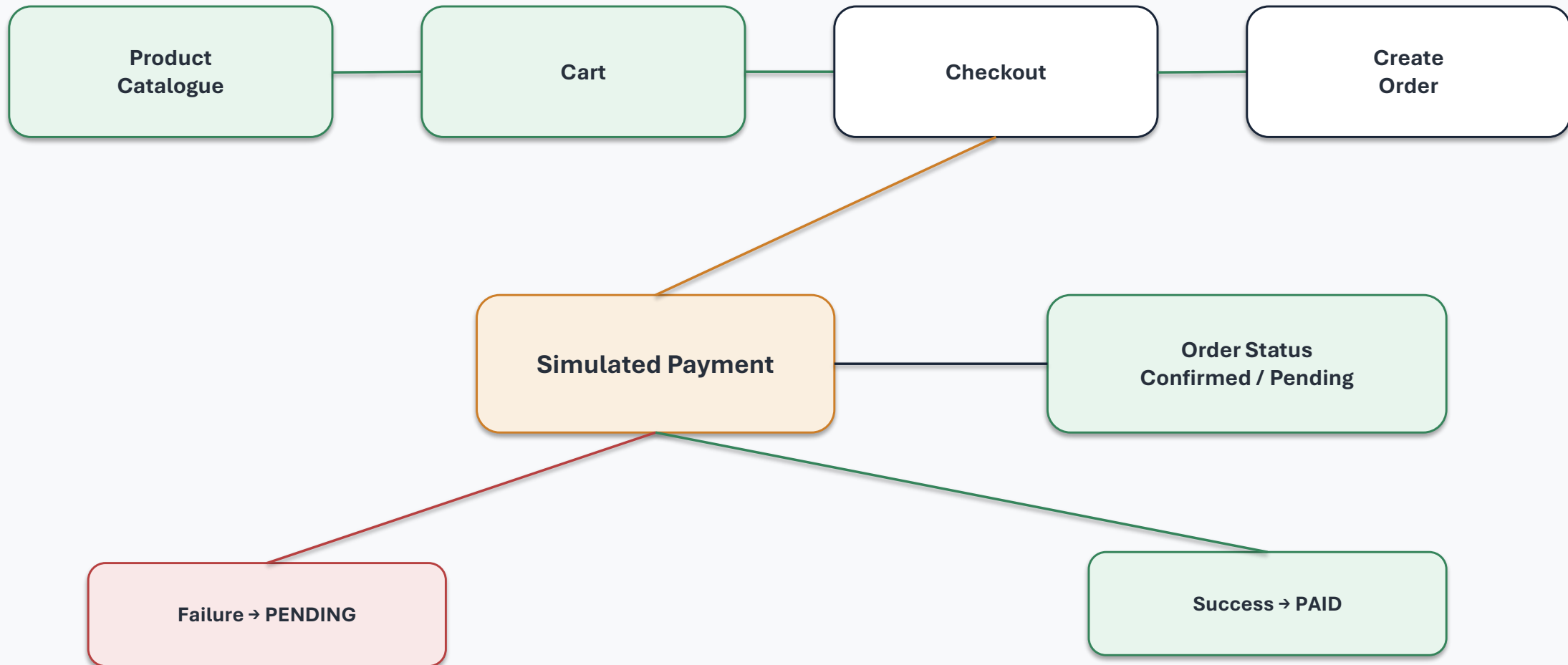


Outcome is persisted and exposed through permit status tracking.

5. Logical View — Marketplace & Payment

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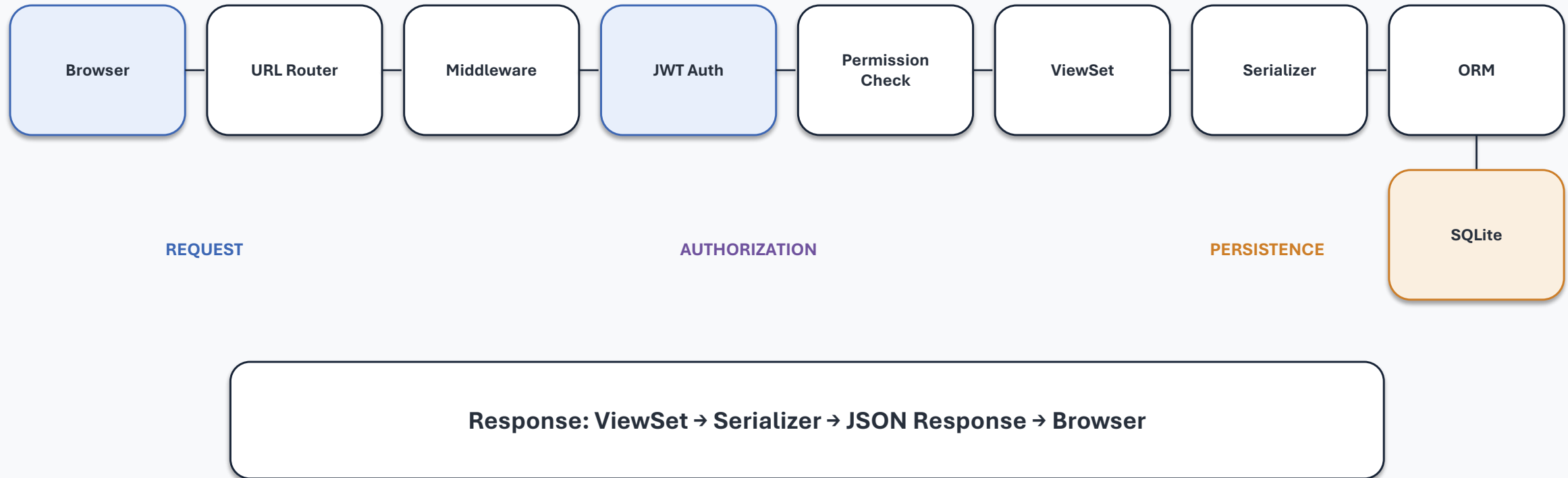
Cart, order and simulated payment remain separate responsibilities



6. Process View — Runtime Request Path

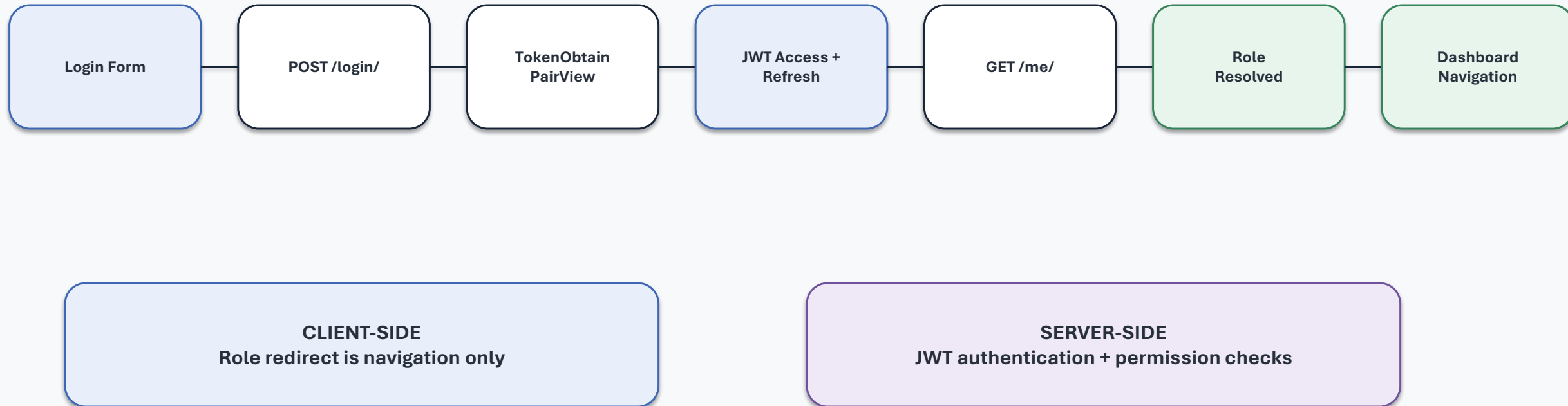
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Browser request through the Django stack



7. Process View — Authentication & RBAC

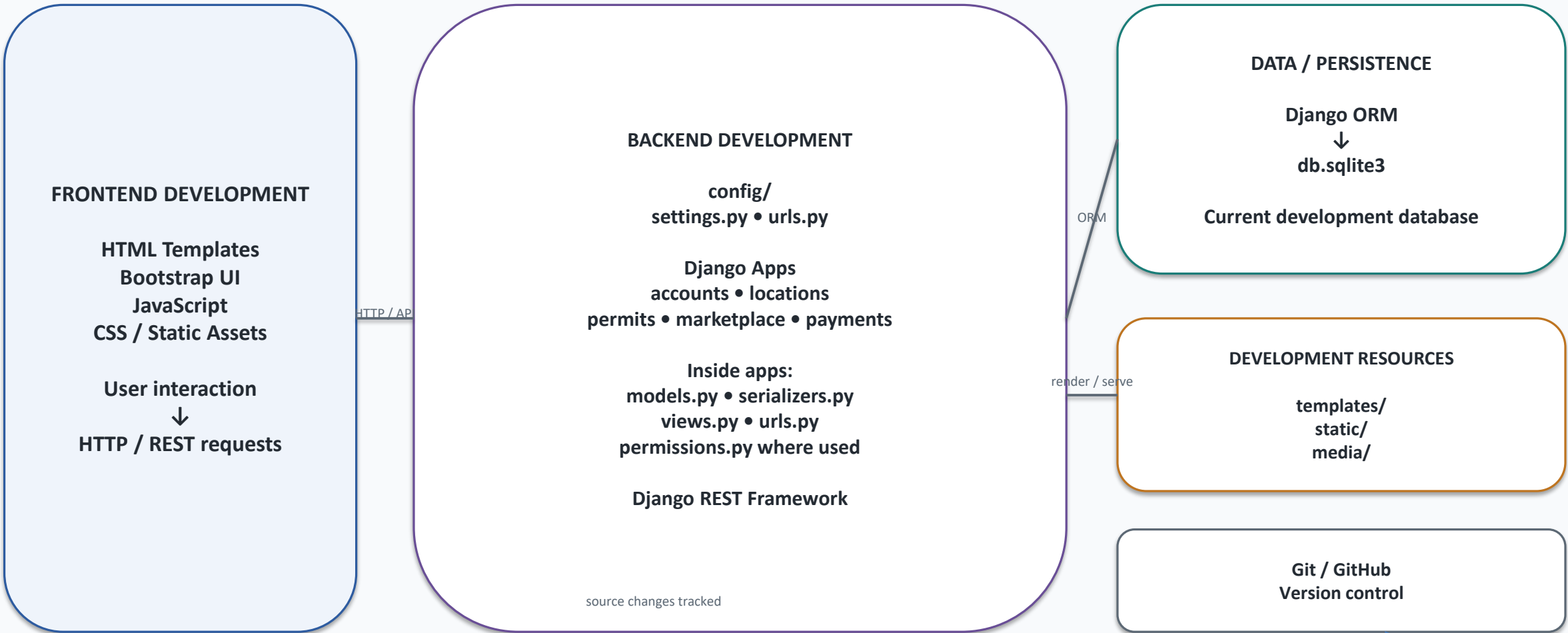
Navigation and authorization are intentionally separated



Every protected API request is authorized at the server boundary.

Development View — Full Development Architecture

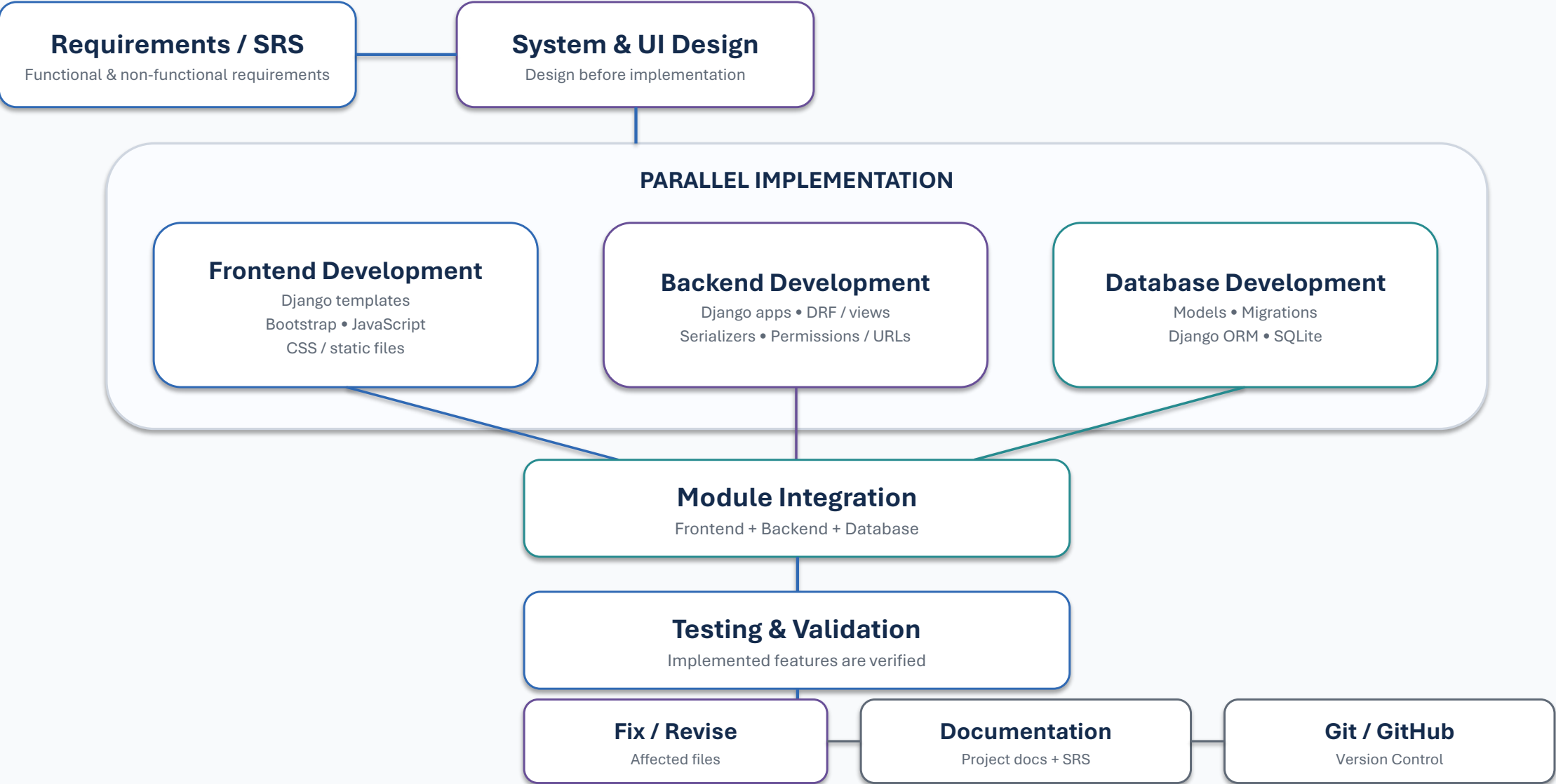
How frontend, backend modules, API integration and persistence are developed together



Development flow: UI implementation → API integration → Django business logic → ORM persistence → testing / iteration. PostgreSQL, Gunicorn and cloud deployment are future only.

Development View — NirmanSathi Development Process

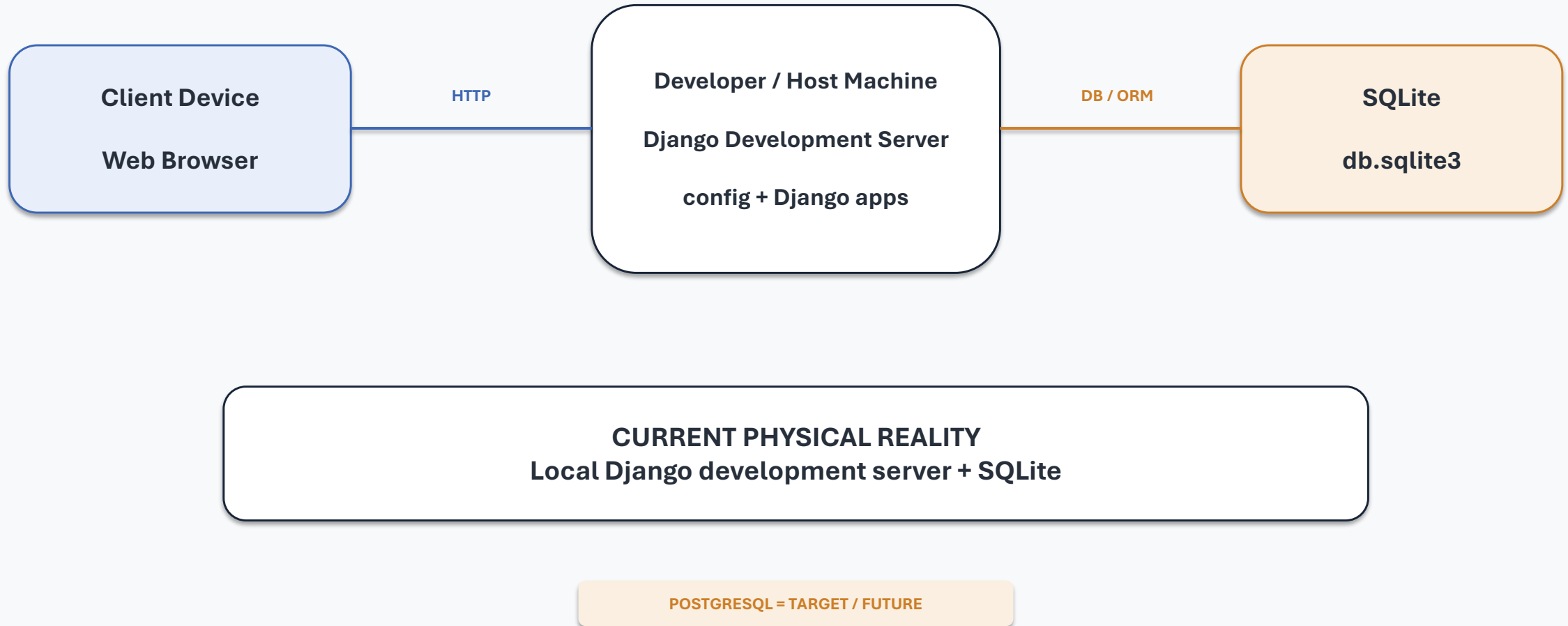
High-level development approach from requirements and design to implementation, integration, testing and version-controlled documentation



Development process view — not runtime request processing or deployment topology

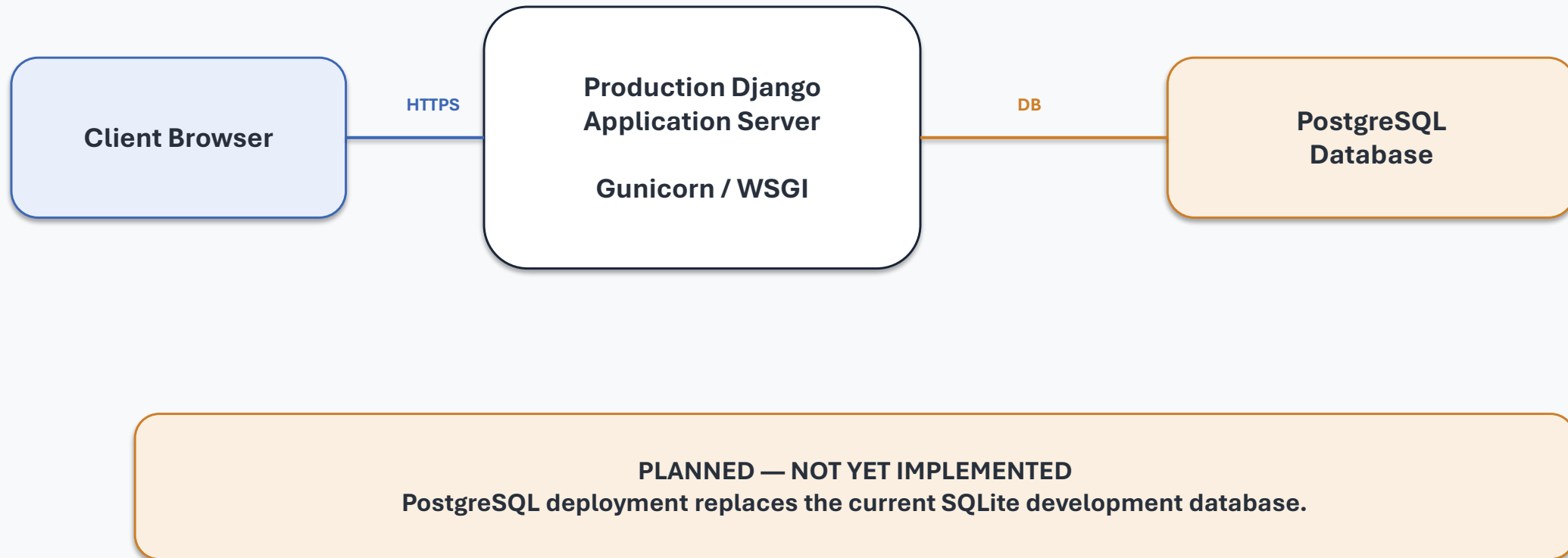
10. Physical View — Current Implementation

Current development topology — not the planned production architecture



11. Physical View — Target Evolution

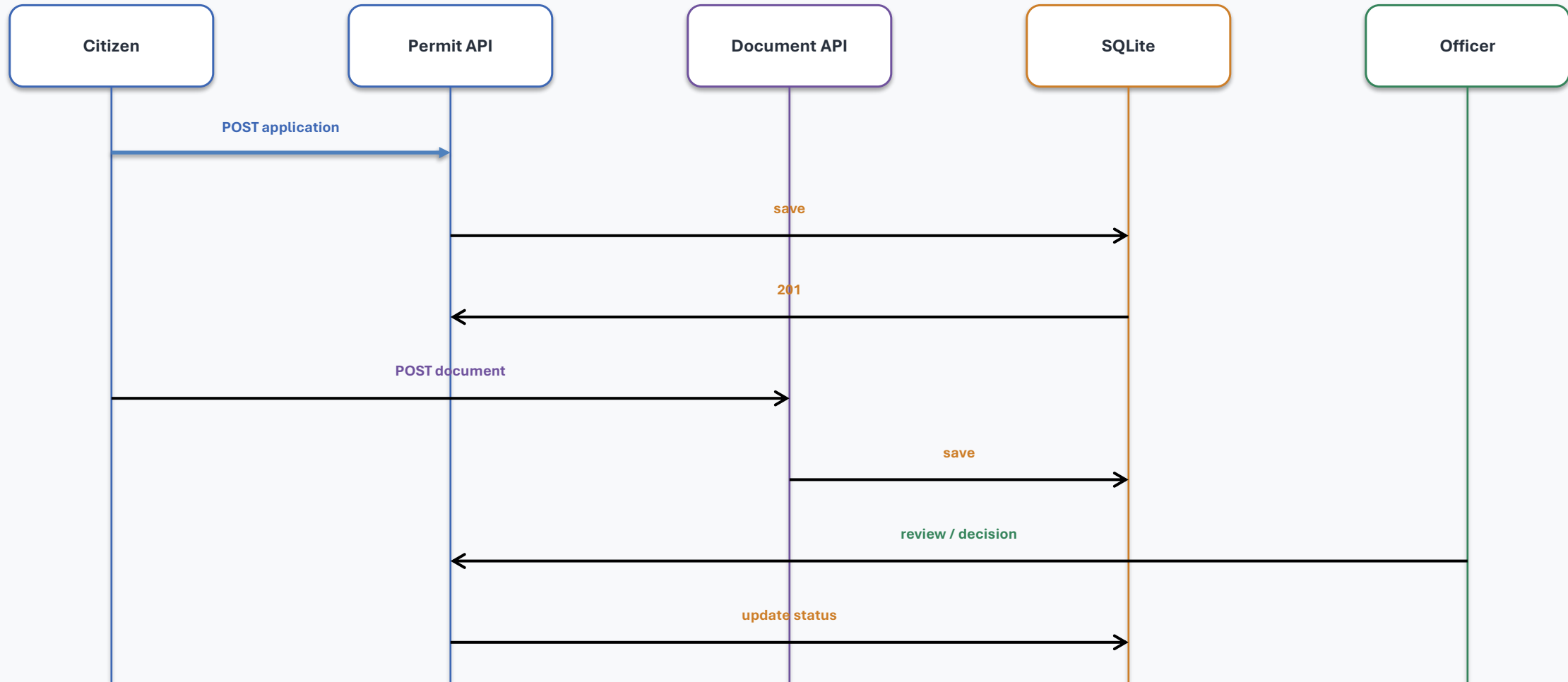
Clearly labelled planned production direction



12. Scenario View — Citizen Permit Submission

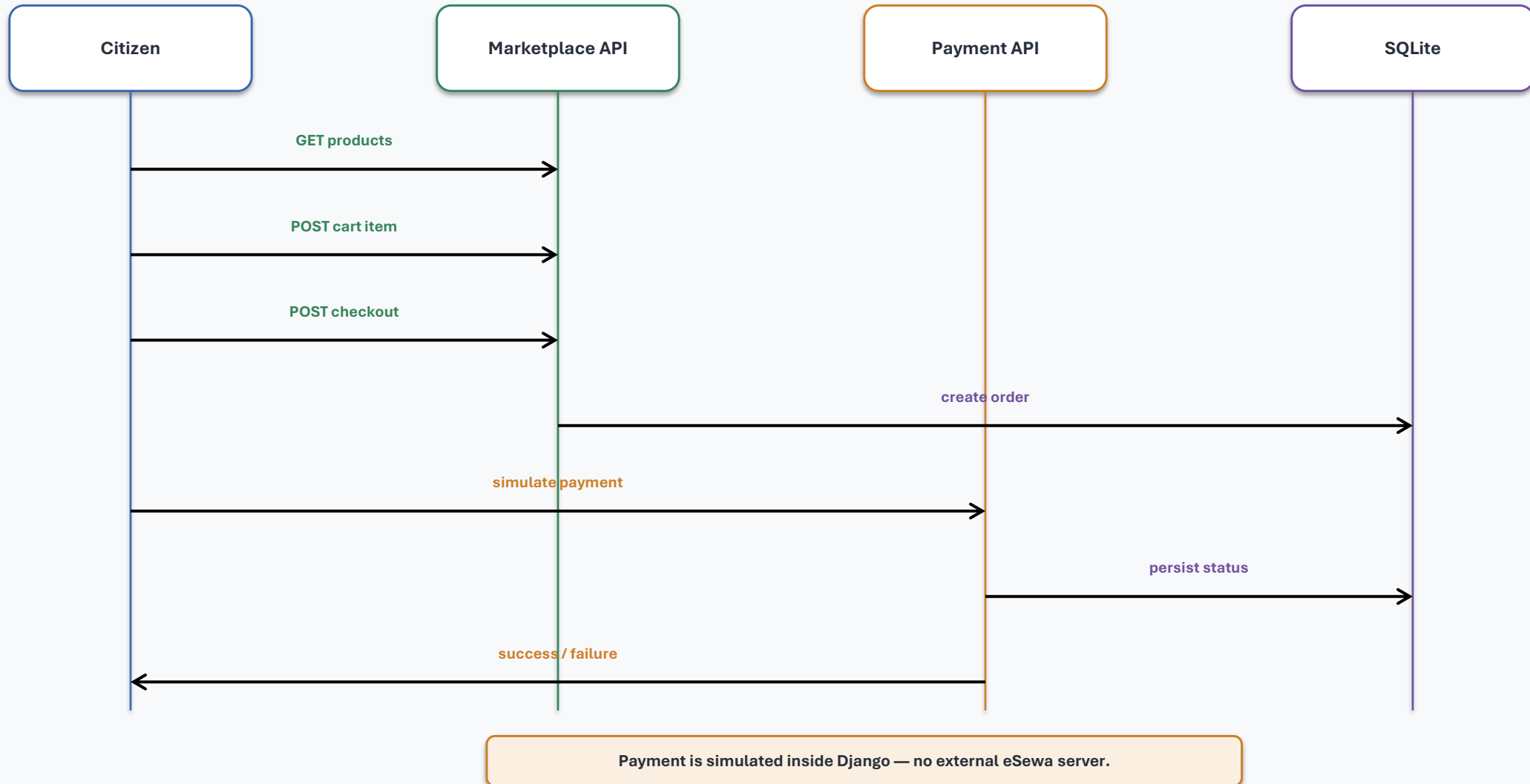
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Sequence: submit → persist → review → update status



13. Scenario View — Marketplace + Simulated Payment

Sequence: cart → checkout → payment → order state



14. Architecture Summary

The four views describe the same system from different perspectives

Logical

Business modules, RBAC, permit lifecycle, marketplace, cart, order and simulated payment.

Process

HTTP request → middleware → JWT → permission → ViewSet → serializer → ORM → SQLite.

Development

Actual Django apps: accounts, locations, permits, marketplace and payments.

Physical

Current: browser → Django development server → SQLite. PostgreSQL is planned.

+1 Scenario

Permit submission and marketplace/payment sequences validate the architecture.